Name of Student: Rikisha Bhattacharya	**PRN: 25070122155**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 10	
TITLE: Abstract Classes and Methods	

Problem Statement

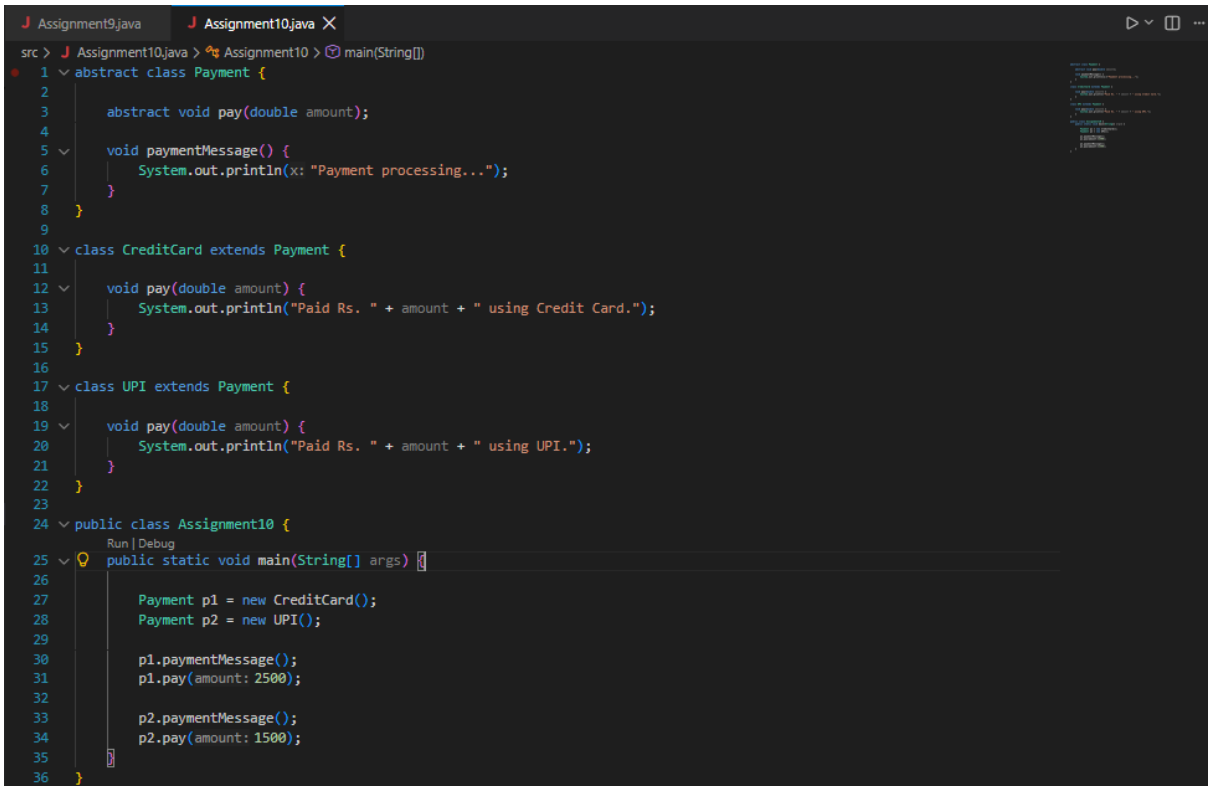
Write a Java program to demonstrate the use of abstract classes and abstract methods

Learning Objectives

To implement abstraction using abstract classes and abstract methods.

1.Create an abstract Payment class and implement Credit Card and UPI payment classes.

Programme



Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                    powershell

PS C:\Users\RIKISHA\PROGRAMMING\Java> cd src
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac Assignment10.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java Assignment10
Payment processing...
Paid Rs. 2500.0 using Credit Card.
Payment processing...
Paid Rs. 1500.0 using UPI.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

2. Create an abstract FoodOrder class with an abstract method calculateBill(). Implement two subclasses, DineInOrder and TakeAwayOrder, to calculate and display the total bill based on the order type.

Programme

Assignment9.java | Assignment10.java

src > Assignment10.java > TakeAwayOrder

```java
abstract class FoodOrder {
    double amount;
    FoodOrder(double amount) {
        this.amount = amount;
    }
    abstract void calculateBill();
}

class DineInOrder extends FoodOrder {

    DineInOrder(double amount) {
        super(amount);
    }

    void calculateBill() {
        double total = amount + 100;
        System.out.println("Dine-In Bill: Rs. " + total);
    }
}

class TakeAwayOrder extends FoodOrder {

    TakeAwayOrder(double amount) {
        super(amount);
    }
    void calculateBill() {
        double total = amount + 50;
        System.out.println("Take-Away Bill: Rs. " + total);
    }
}
public class Assignment10 {
    Run | Debug
    public static void main(String[] args) {

        FoodOrder order1 = new DineInOrder(amount: 500);
        FoodOrder order2 = new TakeAwayOrder(amount: 500);

        order1.calculateBill();
        order2.calculateBill();
    }
}
```

Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                    Debug: Assignment10

PS C:\Users\RIKISHA\PROGRAMMING\Java> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=
localhost:61795' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RIKISHA\PROGRAMMING\Java\bin' 'Assignment10'
Dine-In Bill: Rs. 600.0
Take-Away Bill: Rs. 550.0
PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

Conclusion

This assignment helped us understand abstract classes and abstract methods in Java. We learned how abstract methods are implemented by subclasses to provide specific functionality.

The program demonstrates abstraction, inheritance, and method overriding in Java.

GitHub Screenshot